Toward the Optimal Stimulus to Elicit the Frequency-Following Response

Laura Jacxsens,[1,2,3] Natàlia Gorina-Careta,[4,5,6] Alejandro Mondéjar-Segovia,[4,5] Sonia Arenillas-Alcón,[4,5,6] Lana Biot,[1,2] Emilie Cardon,[1,2] Vincent Van Rompaey,[1,2] Willem De Hertogh,[3] Marc J. W. Lammers,[1,2] and Carles Escera[4,5,6]

Objectives: The frequency-following response (FFR) is gaining momentum to investigate central auditory processing in both audiological sciences and cognitive neuroscience. In previous research, FFRs have been most often elicited by the /da/ stimulus. Recently, we introduced the diphthong /oa/, particularly for infant studies, as, in addition to underpinning fundamental frequency encoding, it provides the underpinnings of temporal fine structure encoding. However, both the /da/ and /oa/ stimuli have their limitations: the /da/ stimulus has been shown to be less effective for studying the temporal fine structure, and the /oa/ stimulus yields a smaller response. In this study, we tested the effects of manipulating vowel order and coarticulation of stimuli as factors that may affect FFR amplitudes.

Design: We conducted a multisite study involving 34 normal-hearing adults (aged 18 to 40) at Antwerp University Hospital and the University of Barcelona. Four different stimuli (/dao/, /doa/, /ao/, and /oa/) designed and equated for intensity were presented in three blocks of 1000 trials in randomized block order. The main FFR parameters were the F_0 and F_1 response amplitude for the /o/ and /a/ sections of the stimuli.

Results: The F_0 amplitude was largest for the /ao/ stimulus and smallest for the /oa/ stimulus in both the /a/ and /o/ vowel segments. In contrast, the /oa/ stimulus evoked significantly higher F_1 amplitudes than all other stimuli across both segments, showing an inverse pattern relative to F_0. The neural response delay was significantly longer for /ao/ compared with the other stimuli. Pre-stimulus root mean square amplitude did not differ across stimuli or sites, indicating consistent baseline neural activity. These amplitude findings were mirrored by significant effects in signal to noise ratio, reinforcing the robustness of the stimulus-driven differences. While a significant Stimulus × Research center interaction was found for F_0 and F_1 amplitudes in the /a/ segment, no interaction effects emerged for the remaining eight FFR parameters, supporting the overall reproducibility and robustness of stimulus effects across centers.

Conclusions: Both amplitude and signal to noise ratio findings confirm that the F_0 response is influenced by the frequency structure of the eliciting vowels and by the preceding vowels and consonants. The choice of stimulus should align with the study's goals: the /ao/ stimulus is ideal for FFR_{ENV} studies, the /oa/ stimulus for FFR_{TFS} studies, and the /dao/ stimulus for studies examining both, offering strong responses for both the FFR_{ENV} and the FFR_{TFS}. The /dao/ stimulus offers the additional advantage that it includes the /da/ stimulus segment, facilitating comparisons with prior research.

Key words: Envelope-following response, Hearing screening, Speech auditory brainstem response, Speech perception, Voice pitch, Vowel formants.

(Ear & Hearing XXX;XX;00–00)

[1]Department of Otorhinolaryngology, Head and Neck Surgery, Antwerp University Hospital, Edegem, Belgium; [2]Faculty of Medicine and Health Sciences, Department of Translational Neurosciences, University of Antwerp, Antwerp, Belgium; [3]Faculty of Medicine and Health Sciences, Department of Rehabilitation Sciences and Physiotherapy, University of Antwerp, Antwerp, Belgium; [4]Brainlab—Cognitive Neuroscience Research Group, Department of Clinical Psychology and Psychobiology, University of Barcelona, Catalonia, Spain; [5]Institute of Neurosciences, University of Barcelona, Catalonia, Spain; and [6]Institut de Recerca Sant Joan de Déu, Catalonia, Spain.

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and text of this article on the journal's Web site (www.ear-hearing.com).

INTRODUCTION

Speech perception is paramount for human communication, yet our understanding of how the human auditory system encodes speech signals at a neural level is still incomplete (Easwar et al. 2021). An interesting tool for evaluating neural processes underlying speech encoding is the frequency-following response (FFR). The FFR is a unique subtype of auditory evoked potential (AEP) due to its ability to closely mimic the temporal and spectral characteristics of the eliciting auditory stimulus. The FFR reflects synchronous neural phase locking to the spectro-temporal components of the acoustic signal (Coffey et al. 2019; Krizman & Kraus 2019). This response arises from multiple generator sources but is thought to be generated primarily in the auditory midbrain, which is a hub of afferent and efferent activity (Coffey et al. 2019; White-Schwoch et al. 2019). These characteristics make the FFR a valuable tool to help understanding central auditory processing of complex stimuli, such as speech, by offering insights into how acoustic features are encoded in neural aggregates, thus advancing our comprehension of speech perception (Kraus & White-Schwoch 2015). Factors such as context, challenging listening conditions (e.g., noise environments), age, and speech and language impairments also shape these neural sound traces (Gorina-Careta et al. 2022). Moreover, the encoding of the fundamental frequency (F_0) of a sound in the FFR is considerably influenced by past auditory experiences (Musacchia et al. 2007). Notably, musical training has been shown to enhance neural representations for speech, resulting in improved speech intelligibility among individuals with musical training compared with those without (Weiss & Bidelman 2015). These enhancements persist even in older populations following short-term musical training (Fleming et al. 2019).

The most used stimulus to elicit the FFR is the /da/ syllable. In our recent systematic review on the FFR in patients with sensorineural hearing loss, the /da/ stimulus was used in 10 out of the 16 included studies (Jacxsens et al. 2024). Similarly, a review by Lemos et al. (2021) on FFR acquisition parameters in newborns and infants found that /da/ was used in 8 of the 15 studies. The popularity of the /da/ syllable is due to several factors, the first of which is that it is a relatively universal syllable

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

<zdoi; 10.1097/AUD.0000000000001764>

that is included in the phonetic inventories of most European languages (Maddieson 1984). Second, this stimulus consists of both a transient and a sustained periodic portion. The transient onset response to the stop burst of the /d/ sound is similar to the click auditory brainstem response (ABR), followed by the tonal /a/ (Skoe & Kraus 2010). The first portion in both the click ABR and FFR is characterized by the activation of higher brainstem regions such as the lateral lemniscus and inferior colliculus (Lemos et al. 2021). In addition, the /da/ syllable provides clear, robust, and replicable responses (Skoe & Kraus 2010). The relevance of using this stimulus relies on the fact that stop consonants pose great perceptual challenges to clinical populations such as the hearing and learning impaired (Turner et al. 1992; Kraus et al. 1996). Because stop bursts are rapid and low in amplitude compared with vowels, even normal-hearing adults can find it difficult to discriminate between contrastive stop consonants (e.g., "dare" versus "bare") in noisy environments (Skoe & Kraus 2010).

The second most frequently used type of stimulus in FFR studies involves pitch-varying tones, such as Mandarin syllables with differing pitch contours (Krishnan et al. 2005; Wong et al. 2007; Song et al. 2008; Skoe & Kraus 2010). This stimulus type has been extensively studied in different populations such as native and nonnative Mandarin speakers (Krishnan et al. 2005; Xu et al. 2006), and musicians and nonmusicians (Wong et al. 2007; Bidelman et al. 2011).

In recent research, we have introduced the diphthong /oa/ as an alternative stimulus to acquire FFRs, particularly in infant studies (Arenillas-Alcón et al. 2021, 2023; Gorina-Careta et al. 2024). This stimulus was designed because the /da/ stimulus has a short duration of the transition between the consonant /d/ and the vowel /a/, as well as a high frequency of the formants (over 500 Hz) that compose the stimulus (Arenillas-Alcón et al. 2021; Gorina-Careta et al. 2022). This makes the /da/ stimulus less effective for evaluating the temporal fine structure of the FFR (Arenillas-Alcón et al. 2021; Gorina-Careta et al. 2022), as the response shows a spectral power decay with increasing frequency (Picton & Burkard 2012). This is particularly critical for populations like newborns that have an immature encoding of the high-frequency content of sounds (Anderson et al. 2015). In contrast, the /oa/ stimulus is composed of two vowel sections with different formant structure based on relatively lower frequency harmonic components and suitable durations for accurate spectral analyses, and therefore enables a proper assessment of both the F_0 and the temporal fine structure (Arenillas-Alcón et al. 2021; Gorina-Careta et al. 2022).

Nevertheless, the /oa/ stimulus has also demonstrated its limitations. In a recent study by Arenillas-Alcón et al. (2023), FFRs were recorded in newborns using both the /da/ and the /oa/. Stimuli were calibrated and presented at the same intensity. The results indicated that the F_0 response amplitude to the vowel /a/ when preceded by /o/ in the syllable /oa/ was smaller than when /a/ was preceded by the consonant /d/ in the syllable /da/. This was an incidental finding that underscores the necessity for further investigation of these effects. Although originally developed for use in infants, the /oa/ stimulus was also tested in adults in the present study to better understand its acoustic and neural characteristics under optimal recording conditions.

Prior research has examined the influence of surrounding consonants and vowels on the FFR. Choi et al. (2013) studied five vowels presented both as steady-state vowels and within three five-word sentences, with each vowel appearing once per sentence. They demonstrated that the same vowel exhibited variations in F_0 response amplitudes when presented in different words. However, these differences were not systematically studied and may therefore have been due to context-dependent changes in vowel acoustics, natural variations in vowel acoustics from production to production, or the changes in consonant context. In a more recent study, Hodge et al. (2018) investigated the impact of speech-shaped noise as a preceding masker on FFRs elicited by /a/. They reported delayed peak latencies in masked conditions, though the amplitudes were not reported.

Easwar and colleagues have also conducted several studies exploring the effects of phonetic context on the envelope-following response (EFR), a term commonly used to describe FFRs derived from averaged polarities, which reflect the stimulus envelope (F_0) and are therefore equivalent to the FFR_{ENV} (phase-independent response) (Easwar et al. 2021, 2022; Easwar & Chung 2022). In these studies, the context surrounding vowels was manipulated in various ways to explore how these changes influence EFR encoding. In one study, the authors embedded 7 vowels in 4 consonant contexts, creating 28 consonant–vowel–consonant syllables. The findings revealed that the type of vowel significantly impacted the EFR amplitude, whereas consonant context did not (Easwar et al. 2021). In a separate experiment, they manipulated the phonemic context preceding a vowel (/i/) and found that while the context had no effect on EFRs elicited by the first formant (F_1), certain consonants (more specifically /ʃ/ and /m/) attenuated EFRs elicited by the second and higher formants (F_{2+}) under specific conditions (Easwar et al. 2022). In a third study, Easwar and Chung (Easwar & Chung 2022) investigated these effects over a longer timescale (adaptation). They revealed that preceding phoneme sequences reduced EFR amplitude for F_{2+} but only during the initial stimulus presentation, with again no effect observed for F_1. The variability in EFR sensitivity to preceding acoustic context reported across studies has been attributed to differences in both the acoustic characteristics of the context itself and the properties of the target stimulus (Easwar et al. 2022).

The primary objective of the present study was to examine how speech encoding strength varies based on vowel order and coarticulation. This was done with the goal of optimizing stimuli that yield FFRs providing valuable information on both the F_0 and temporal fine structure, because both the /da/ and /oa/ stimuli have shown their limitations. Specifically, we investigated the effects of introducing the consonant /d/ at the beginning of the stimulus and altering the sequence of the /o/ and /a/ vowels. In addition, we aimed to evaluate the generalizability of these responses by conducting the study across two research centers.

MATERIALS AND METHODS

Study Design

This is a cross-sectional, multisite study, which was performed at the Antwerp University Hospital in Belgium and the University of Barcelona in Spain. The study was designed and conducted according to the Declaration of Helsinki (1996), approved by the local ethics committee of the Antwerp University Hospital (identification: B3002022000172), the Ethical Committee of Clinical Research (CEIC) of the Sant Joan de Déu Foundation (IRB reference: PIC-53-17), and the

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

TABLE 1. Characteristics of each of the four speech stimuli

Stimulus							
/oa/		Onset	/o/	Transition	/a/		
	Duration (msec)	10	70	10	70		
	F_0 (Hz)		113	113	113		
	F_1 (Hz)		452	452–678	678		
	F_2 (Hz)		791	791–1017	1017		
/ao/		Onset	/a/	Transition	/o/		
	Duration (msec)	10	70	10	70		
	F_0 (Hz)		113	113	113		
	F_1 (Hz)		678	678–452	452		
	F_2 (Hz)		1017	1017–791	791		
/doa/		Onset	/d/	Transition	/o/	Transition	/a/
	Duration (msec)	10	47	10	70	10	70
	F_0 (Hz)		113	113	113	113	113
	F_1 (Hz)		553–565	565–452	452	452–678	678
	F_2 (Hz)		1438–904	904–791	791	791–1017	1017
/dao/		Onset	/d/	Transition	/a/	Transition	/o/
	Duration (msec)	10	47	10	70	10	70
	F_0 (Hz)		113	113	113	113	113
	F_1 (Hz)		553–565	565–678	678	678–452	452
	F_2 (Hz)		1438–904	904–1017	1017	1017–791	791

Bioethics Committee of the University of Barcelona. The study is reported following the STROBE guideline. Written informed consent was obtained from each participant before starting the experiment.

Participants

Healthy young adults, aged between 18 and 40 years old, were enrolled in both research centers. To reduce variability in language lateralization, individuals who self-identified as left-handed were excluded from the study, given established evidence suggesting differences in language lateralization between left- and right-handed individuals (Bruckert et al. 2021). In addition, participants were excluded if they self-reported habitual use of psychopharmaceutical drugs or medications affecting the central nervous system, drug consumption within the preceding 48 hours, or significant illness on the day of the experiment. To minimize potential effects of musical training on the FFR (Musacchia et al. 2007; Kraus & Chandrasekaran 2010), participants were recruited if they had less than four years of musical training, which must have ended at least five years before data collection.

Participants underwent hearing screening by performing a pure-tone audiometry. In Antwerp, pure-tone audiometry was measured in all participating subjects according to the clinical standards (ISO 8253-1, 2010) using a two-channel Interacoustics AC-40 audiometer in a soundproof booth. Air conduction thresholds were measured at 125 Hz, 250 Hz, 500 Hz, 1 kHz, 2 kHz, 3 kHz, 4 kHz, 6 kHz, and 8 kHz. Participants were required to have pure-tone audiometry thresholds of 20 dB HL or lower at each of these frequencies. In Barcelona, participants' hearing thresholds were also assessed with pure-tone audiometry using ER-3A ABR insert earphones (Etymotic Research, Inc., Elk Grove Village, IL, USA) in an electrically and acoustically shielded room. Mean hearing thresholds were below 20 dB HL for the five test frequencies (250, 500, 1000, 2000, and 4000 Hz) and interaural threshold differences of less than 10 dB in all the participants.

This resulted in the inclusion of 34 participants across both research centers: 20 participants in Antwerp (10 males, mean age = 26.93 years ± 4.05) and 14 participants in Barcelona (6 males, mean age = 28.85 years ± 6.90).

Stimuli

In this study, we used four speech stimuli: /dao/, /doa/, /ao/ and /oa/. Stimuli were designed using the Praat software (Boersma & Weenink 2013). The total duration of both /dao/ and /doa/ was 217 msec, while /ao/ and /oa/ were 160 msec in duration. Each stimulus started with an onset period of 10 msec, followed by a consonant transition of 47 msec during the /d/ section in /dao/ and /doa/. The /o/ and /a/ sections were 70 msec in duration, with a 10 msec transition between individual letters. The F_0 was held constant at 113 Hz throughout the entire duration of each stimulus, with the spectral content ranging from 113 to 1438 Hz. Table 1 provides the detailed spectral characteristics of each stimulus, while Figure 1 illustrates their time-domain waveforms and spectrograms.

Data Acquisition

The procedural methods slightly varied between the two research centers, as data acquisition in Antwerp was conducted in a clinical setting, whereas data acquisition in Barcelona was carried out in a basic research laboratory. These differences were intentional and allowed us to compare whether FFR outcomes remain stable across research-driven and clinically oriented acquisition protocols.

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

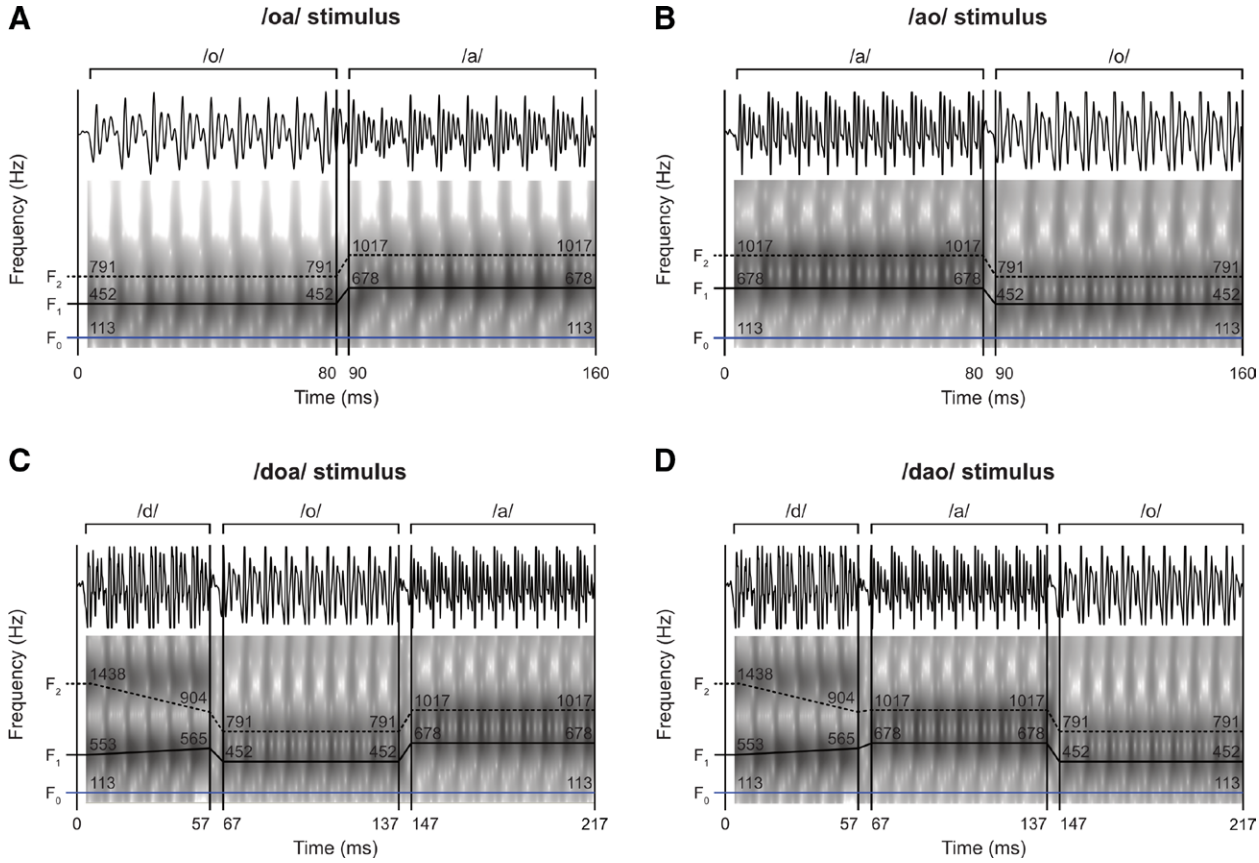

Fig. 1. Time-domain waveforms and spectrograms of the four stimuli. For each stimulus, the upper panel displays the time-domain waveform, and the lower panel shows the corresponding narrow-band spectrogram. The stimulus fundamental frequency (F_0 = 113 Hz) is overlaid in blue. Harmonic contours, as defined in Table 1, are plotted in black: the first harmonic (F1) is shown as a solid bold line, and the second harmonic (F_2) as a dashed line.

In Barcelona, stimuli were presented binaurally (diotically) at an intensity of 75 dB SPL. This setup reflects a typical basic research approach, as binaural stimulation is commonly used to elicit stronger FFRs in studies investigating auditory plasticity, musicianship, or bilingualism (Font-Alaminos et al. 2021). In contrast, in Antwerp, we used a clinical protocol in which stimuli were presented monaurally to the right ear at 70 dB SPL. Although stimulation of one ear was sufficient for this study, this approach was selected to align with clinical standards for ear-specific auditory assessment. In both research centers, stimuli were presented in alternating polarities via ER-3A ultra-shielded insert earphones. The spectral content up to the second formant (F_2) of the stimuli was fully compatible with the ER-3A earphone frequency response, ensuring accurate stimulus delivery.

In Antwerp, FFRs were collected using the "SmartEP" platform connected to a "Duet" amplifier, which includes the "complex Auditory Brainstem Response" (cABR) and "Advanced Hearing Research" modules (Intelligent Hearing Systems [IHS], Miami, FL, USA). Responses were recorded with disposable snap Ag/AgCl electrodes placed in a vertical montage (active electrode at AFz; reference electrode at the right mastoid; ground at the lower forehead). The skin was prepared by use of a Nuprep gel to lower the skin impedance, which had to be below 5 kOhm and inter-electrode impedance had to be below 2 kOhm. Any electrical activity exceeding ±30 μV was automatically rejected. During acquisition, an online bandpass filter of 30-1500 Hz was applied, and signals were sampled at a rate of 40 kHz.

In Barcelona, continuous EEG recording was acquired with Neuroscan 4.4 software and Neuroscan SynAmps RT amplifier (NeuroScan, Compumedics, Charlotte, NC, USA). EEG was recorded from 32 scalp Ag/AgCl electrodes mounted in a Quick-Cap (Compumedics) according to the international 10 to 20 system. Two additional electrodes were positioned at the mastoids, with the right mastoid serving as the online reference and the ground located between Fz and Fpz. The EEG was digitized at a 20 kHz sampling rate, with online bandpass filtering from 0.05 to 3000 Hz. Stimuli were delivered binaurally at 75 dB SPL using ER-3A insert earphones.

Both in Antwerp and Barcelona, all measurements took place in an electrically and acoustically shielded room. Participants were seated comfortably in a reclined chair, and the light was subdued. During the measurements, participants were instructed to relax, but stay awake, while watching a silent movie (Planet Earth II for participants in Antwerp, movie of choice in Barcelona). To ensure that subjects remained awake, regular breaks were taken in between blocks. Three blocks of 1000 responses to each stimulus were recorded, with each block lasting approximately 5 to 6 min. The order of the blocks of the four stimuli was randomized and counterbalanced across participants. The total mean duration of the experimental session, including electrode attachment and detachment, was approximately 90 min.

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

Data Processing and Analysis

Data were bandpass filtered from 80 to 1500 Hz. For data collected in Antwerp, data were epoched online and averaged from −40.95 msec (pre-stimulus period) to 100 msec after the end of the stimulus (260 msec for /oa/ and /ao/, 317 msec for /doa/ and /dao/). For data collected in Barcelona, data analysis was performed offline using EEGlab v.7 toolbox (Delorme & Makeig 2004) running under Matlab v.2012a. The continuous recordings extracted from the FPz electrode were filtered offline with a bandpass Kaiser window FIR filter from 70 to 1500 Hz and epoched from 40 msec before the stimulus onset to 250 msec after the stimulus onset. Epochs for each of the stimuli were sorted separately. Trials with activity greater than 35 μV were removed from any further analysis and remaining epochs were baseline corrected to a 40 msec interval preceding the sound onset.

The neural responses evoked by the two opposite stimulus polarities were averaged [(Condensation + Rarefaction)/2], to emphasize the encoding of the envelope of the stimulus (F_0) and reduce any phase-dependent response, such as the cochlear microphonic response. This resulted in the FFR_{ENV}. To emphasize the FFR components associated with the encoding of the temporal fine structure, such as the first formant (F_1), while reducing the impact of envelope-related activity, the neural responses to alternating polarities were subtracted [(Condensation − Rarefaction)/2], resulting in the FFR_{TFS} (Aiken & Picton 2008; Krizman & Kraus 2019). Custom Matlab scripts developed at the University of Barcelona and utilized in similar analyses in earlier studies were implemented to compute all parameters (Arenillas-Alcón et al. 2021, 2023; Gorina-Careta et al. 2024). In addition, a further custom Matlab script was implemented to ensure that the Neuroscan data could be processed alongside the IHS data using the same script, to ensure comparability of the FFR outcomes. Several FFR parameters were computed as described later.

Pre-Stimulus Root Mean Square Amplitude • The root mean square (RMS) of the pre-stimulus period was taken as a measure of the overall magnitude of neural activity over time and used to eliminate electrophysiological differences in the pre-stimulus region (Liu et al. 2015; White-Schwoch et al. 2015; Ribas-Prats et al. 2019, 2022; Arenillas-Alcón et al. 2023; Gorina-Careta et al. 2024). It was calculated by first squaring each data point of the pre-stimulus region of the neural response (from −40 to 0 msec), followed by calculating the mean of the squared values, and finally calculating the square root of the obtained average.

Neural Lag • Neural lag was identified as an indicator of the neural transmission delay of the auditory system, as this value provides insight into the time elapsed between the reception of the stimulus at the cochlea and the onset of the neural phase locking (Liu et al. 2015; Ribas-Prats et al. 2019, 2022; Arenillas-Alcón et al. 2021, 2023; Gorina-Careta et al. 2024). It was calculated by performing a cross-correlation analysis between the auditory stimulus and the neural response. The correlation value at each time lag within a 3 to 13 msec time window was obtained. The neural lag was then determined by selecting the time lag corresponding to the highest cross-correlation value within this time window (Arenillas-Alcón et al. 2023).

Spectral Amplitude at F_0 for Both the /a/ and /o/ Sections of the Stimuli • To determine the strength of the response, the spectral amplitude at F_0 (113 Hz) was considered as a measure of the neural phase-locking magnitude at the specific selected frequency (White-Schwoch et al. 2015; Ribas-Prats et al. 2019, 2022; Arenillas-Alcón et al. 2021, 2023; Gorina-Careta et al. 2024). To examine the FFR in the frequency domain, we computed a fast Fourier Transform (Hanning windowed) separately for the response sections corresponding to the /a/ and /o/ vowels of the four different stimuli (see Table 1 for an overview of stimulus segmentation). Each region was adjusted for each participant to account for the individual neural lag. The use of a Hanning window minimized spectral leakage due to signal truncation. To estimate the amplitude of the neural response more robustly, we calculated the mean spectral amplitude within a ±5 Hz window centered at the stimulus F_0 frequency. This approach captures the spread of energy due to the windowing function and accounts for minor inter-individual variability in frequency representation.

SNR at F_0 Computed for the /o/ and /a/ Sections of the Stimulus • The SNR at the F_0 peak was analyzed to obtain an estimation of the relative spectral magnitude of the response, considering not only the amplitude value of the signal at the frequency peak of interest (113 Hz) but also the noise level at the surrounding frequencies. To compute the SNR, we divided the signal power—defined as the mean spectral power within a ±5 Hz frequency bin centered at 113 Hz (i.e., 108 to 118 Hz)—by the noise power, defined as the mean spectral power of two 28 Hz-wide flanking bins centered approximately ±57 Hz from the peak (i.e., centered at 56 Hz and 170 Hz, covering 42 to 70 Hz and 156 to 184 Hz, respectively). This spacing corresponds to half the F_0 and was chosen to minimize contamination from the peak's spectral slope. The SNR was expressed in dB as: SNR = $10 \times \log_{10}$ (Signal power/Noise power) (Ribas-Prats et al. 2019, 2022; Arenillas-Alcón et al. 2021). The SNR was computed for the response sections that corresponded to the /a/ and /o/ parts of the different stimuli.

Spectral Amplitudes at F_1 for Both the /a/ and /o/ Sections of the Stimuli • To assess spectral amplitudes at the specific spectral peaks for the stimulus F_1 (452 Hz [/o/] and 678 Hz [/a/]), the neural responses corresponding to the /o/ section and the /a/ steady section of the stimuli were individually analyzed. As with the F_0 analysis, a Hanning window was applied to each response segment before the fast Fourier Transform, and the respective amplitudes within a ±5 Hz window centered at the peak frequencies corresponding to the vowel formant centers were extracted. This method ensured that spectral energy related to the true response frequency was captured while minimizing contributions from spectral leakage and noise.

SNR at F_1 Computed for the /o/ and /a/ Sections of the Stimulus • This parameter was computed for the /o/ and /a/ sections of the stimuli in the same manner as described for F_0. Specifically, SNRs were calculated at the spectral peaks corresponding to the F_1 frequencies of the stimuli (452 and 678 Hz), using ±5 Hz signal windows and flanking 28 Hz noise windows centered at ±57 Hz from each frequency of interest.

Statistical Analysis

Statistical analyses were conducted using IBM SPSS Statistics (version 27, IBM Corp., Armonk, NY). Descriptive

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

statistics, including the mean, SD, median, first quartile (Q1), third quartile (Q3), interquartile range, minimum, and maximum values, were calculated for each parameter acquired from the different stimuli.

To determine whether a statistically significant response was present at the frequency of interest, an *F* test was applied following Encina-Llamas et al. (2019) and Dobie and Wilson (1996). The *F* ratio was defined as the power in a 10-Hz window centered on the frequency of interest, divided by the average power in two 28-Hz noise windows on either side. Because our SNR values were log-transformed (dB), we inverted them to obtain linear *F* ratios. The *F* distribution had degrees of freedom $df_1 = 20$ (signal) and $df_2 = 112$ (noise). The *F* test was defined to be positive if $p \leq 0.01$ (*F* critical value ≥2.0460, SNR ≥3.10 dB).

To analyze the effects of stimulus and research center, linear mixed-effect models were used. Each FFR parameter was analyzed separately with stimulus (4 levels: /doa/, /dao/, /oa/, and /ao/) research center (two levels: University of Barcelona and Antwerp University Hospital), and their interaction (Stimulus × Research center) entered as fixed factors. The subject was included as a random intercept to account for repeated measures within individuals. This approach allowed us to statistically regress out the effect of site, enabling a more accurate comparison between stimulus conditions.

Type III tests of fixed effects were used to evaluate the significance of the main and interaction effects. For models where a significant Stimulus × Research center interaction was observed, additional post hoc comparisons using estimated marginal means were conducted to examine stimulus effects within each site separately. Where the main effect of the stimulus was significant, Bonferroni-adjusted post hoc pairwise comparisons were performed to explore differences between stimulus conditions. To control for multiple testing across the FFR parameters analyzed, a Bonferroni correction was applied to the overall model-level significance threshold.

RESULTS

The neural responses to the fundamental frequency (F_0) in the vowel segments /o/ and /a/ were analyzed. All detailed descriptive statistics from all analyzed parameters can be found in Section A in Supplemental Digital Content, https://links.lww.com/EANDH/B792.

Presence of Frequency-Specific Responses

We applied an *F* test to evaluate whether a response was present at the frequency of interest. *F* tests were first applied to the mean SNRs of each stimulus segment (/o/ and /a/ section) of the four different stimuli. All eight means showed significant responses at the group level ($p \leq 0.01$). This suggests that, on average, all conditions elicited a robust frequency-specific response.

We also applied the *F* test at the individual level. In most stimuli, more than 90% of participants exhibited significant *F* test results. The only exception was again the /a/ segment of the /oa/ stimulus, where 85.3% of participants showed a significant response. These results confirm that most participants exhibited a statistically significant response across conditions (see Section B in Supplemental Digital Content, https://links.lww.com/EANDH/B792). It is important to note that all individual data points were retained in the statistical analyses, regardless of whether they reached significance in the *F* test.

Pre-Stimulus RMS Amplitude

We did not observe a significant main effect of stimulus ($F_{3,96} = 1.40$, $p = 1.000$) or research center ($F_{1,32} = 4.55$, $p = 0.407$) on pre-stimulus RMS amplitude. In addition, the interaction between stimulus and research center was not significant ($F_{3,96} = 1.70$, $p = 1.000$), indicating that the relationship between stimulus type and pre-stimulus RMS amplitude was consistent across the two research sites.

A table with all linear mixed-effects model results for all FFR parameters can be found in Section C in Supplemental Digital Content, https://links.lww.com/EANDH/B792.

Neural Transmission Delay

There was a significant main effect of the stimulus on the neural lag ($F_{3,96.01} = 9.69$, $p = < 0.001$). Post hoc pairwise comparisons with Bonferroni correction revealed that /ao/ elicited a significantly longer neural lag compared with all other stimuli (/doa/: $p < 0.001$, /dao/: $p = 0.002$, /oa/: $p < 0.001$). No other pairwise comparison between /doa/, /dao/, and /oa/ reached significance, indicating that the increased lag in /ao/ was the primary driver of the main effect.

In contrast, there was no significant main effect of the research center ($F_{1,32.02} = 1.61$, $p = 1.000$), nor was there a significant interaction between stimulus and research center ($F_{3,96.01} = 2.26$, $p = 0.859$), suggesting that the pattern of neural lag across stimuli was consistent across both sites.

F_0 Response – /a/ Section

Figure 2 shows the grand-averaged FFR_{ENV} waveforms, frequency spectra, and boxplots of spectral amplitudes at F_0 for the /a/ section.

The linear mixed-effects model revealed significant main effects of stimulus ($F_{3,96} = 14.45$, $p < 0.001$) and research center ($F_{1,32} = 17.26$, $p = 0.002$) on F_0 spectral amplitude. Post hoc comparisons revealed that /ao/ elicited significantly larger spectral amplitudes than all other stimuli (/doa/: $p < 0.001$, /dao/: $p = 0.001$, /oa/: $p < 0.001$). Regarding the research center, amplitude values measured at the Antwerp University Hospital were systematically lower than those measured at the University of Barcelona (estimated difference = −0.109 μV, $p < 0.001$).

In addition, there was a significant interaction between stimulus and research center ($F_{3,96} = 5.07$, $p = 0.026$), suggesting that the effect of stimulus on F_0 amplitude varied between the two sites. Post hoc analyses revealed that the F_0 amplitude significantly differed between stimuli at the University of Barcelona ($F_{3,96} = 15.56$, $p < 0.001$), with /ao/ eliciting significantly larger amplitudes than all other stimuli (all $p < 0.001$). In contrast, no significant stimulus effects were observed at the Antwerp University Hospital ($F_{3,96} = 1.49$, $p = 0.223$). This confirms that the stimulus-driven differences in F_0 response amplitude were site-dependent. A summary of estimated marginal means and significant pairwise comparisons is provided in Section D in Supplemental Digital Content, https://links.lww.com/EANDH/B792.

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

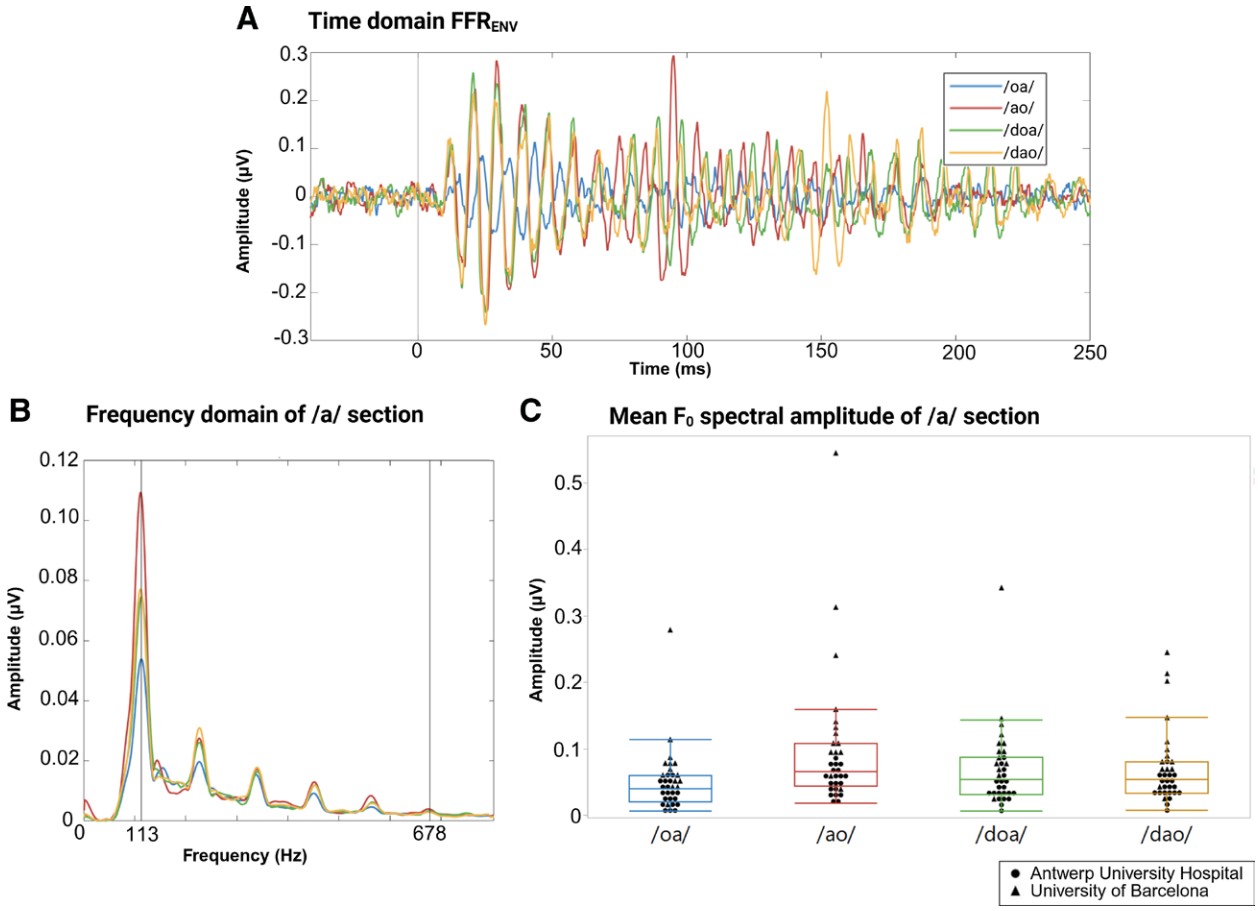

Fig. 2. Results for the FFR$_{ENV}$ for /a/ section. A, Grand-averaged time-domain waveform of the FFR$_{ENV}$ elicited by the four different stimuli. B, FFR$_{ENV}$ frequency spectra extracted from the equivalent /a/ analyzed section of the neural response. C, Boxplots of the mean F$_0$ spectral amplitude from the equivalent /a/ analyzed section of the neural response for each stimulus.

For the SNR at F$_0$, significant main effects were found for both stimulus ($F_{3,96} = 18.07$, $p < 0.001$) and research center ($F_{1,32} = 30.05$, $p < 0.001$). Post hoc comparisons revealed that /ao/ elicited significantly higher SNR values than all other stimuli (/doa/: $p < 0.001$, /dao/: $p < 0.001$, /oa/: $p = 0.004$), while /oa/ showed significantly lower SNR than all others. No significant interaction was observed ($F_{3,96} = 1.20$, $p = 1.000$).

F$_0$ Response – /o/ Section

Figure 3A shows the spectral representations of the FFR$_{ENV}$ from the /o/ section of each stimulus, and Figure 3B presents boxplots of spectral amplitudes at F$_0$.

The analysis revealed significant main effects of both stimulus ($F_{3,96.01} = 7.42$, $p = 0.002$) and research center ($F_{1,32} = 39.50$, $p < 0.001$) on the F$_0$ response amplitude of the /o/ section. Post hoc tests showed that /oa/ yielded significantly lower spectral amplitudes compared with /doa/ ($p = 0.002$) and /ao/ ($p < 0.001$). Again, amplitudes were lower at Antwerp University (estimated difference = −0.076 µV, $p < 0.001$). No significant Stimulus × Research center interaction was observed ($F_{3,96.01} = 2.25$, $p = 0.877$).

For the SNR at F$_0$, the analysis showed significant main effects of stimulus ($F_{3,69.01} = 5.46$, $p = 0.017$) and research center ($F_{1,32.01} = 21.24$, $p < 0.001$). Post hoc comparisons revealed that /oa/ elicited significantly lower SNR values than /doa/ ($p = 0.019$) and /ao/ ($p = 0.005$). No significant interaction was found ($F_{3,96.01} = 2.36$, $p = 0.766$).

F$_1$ Response – /a/ Section

Figures 4A, B show the grand-averaged FFR$_{TFS}$ waveforms and spectral representations, Figure 4C presents boxplots of spectral amplitudes at F$_1$ for the /a/ section.

There were significant main effects of both stimulus ($F_{3,96.01} = 38.22$, $p < 0.001$) and research center ($F_{1,32} = 70.72$, $p < 0.001$) on the F$_1$ response amplitude. Pairwise comparison revealed that the F$_1$ response amplitude of /oa/ was significantly larger than that of /doa/ ($p < 0.001$), /dao/ ($p < 0.001$), and /ao/ ($p < 0.001$). Moreover, F$_1$ response amplitudes measured at Antwerp University Hospital were significantly smaller than those measured at the University of Barcelona (estimated difference = −0.010 µV, $p < 0.001$).

The model also showed a significant Stimulus × Research center interaction ($F_{3,96.01} = 14.67$, $p < 0.001$), indicating that the effect of stimulus on F$_1$ amplitude differed between the two sites.

To explore this interaction, additional post hoc comparisons were conducted within each site.

In Barcelona, F$_1$ amplitude significantly differed according to the used stimulus ($F_{3,96.01} = 43.65$, $p < 0.001$). The /oa/ elicited significantly higher amplitudes than all other stimuli (all $p < 0.001$), and /ao/ showed significantly higher amplitudes than /doa/ ($p = 0.030$).

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

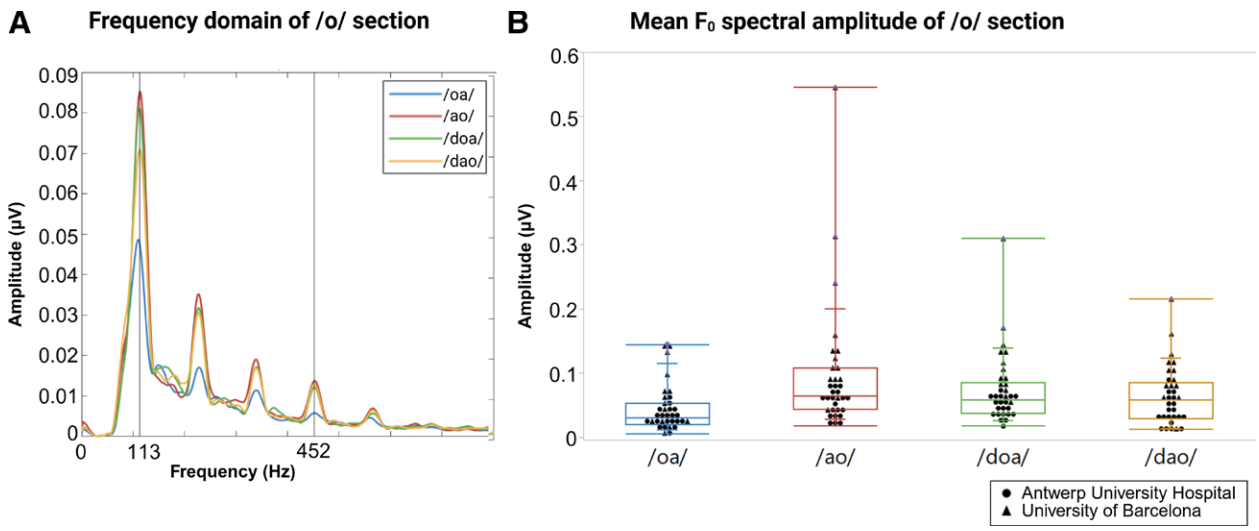

Fig. 3. Results for the FFR_{ENV} for /o/ section. A, FFR_{ENV} frequency spectra extracted from the equivalent /o/ analyzed section of the neural response. B, Boxplots of the mean F_0 spectral amplitude from the equivalent /o/ analyzed section of the neural response for each stimulus.

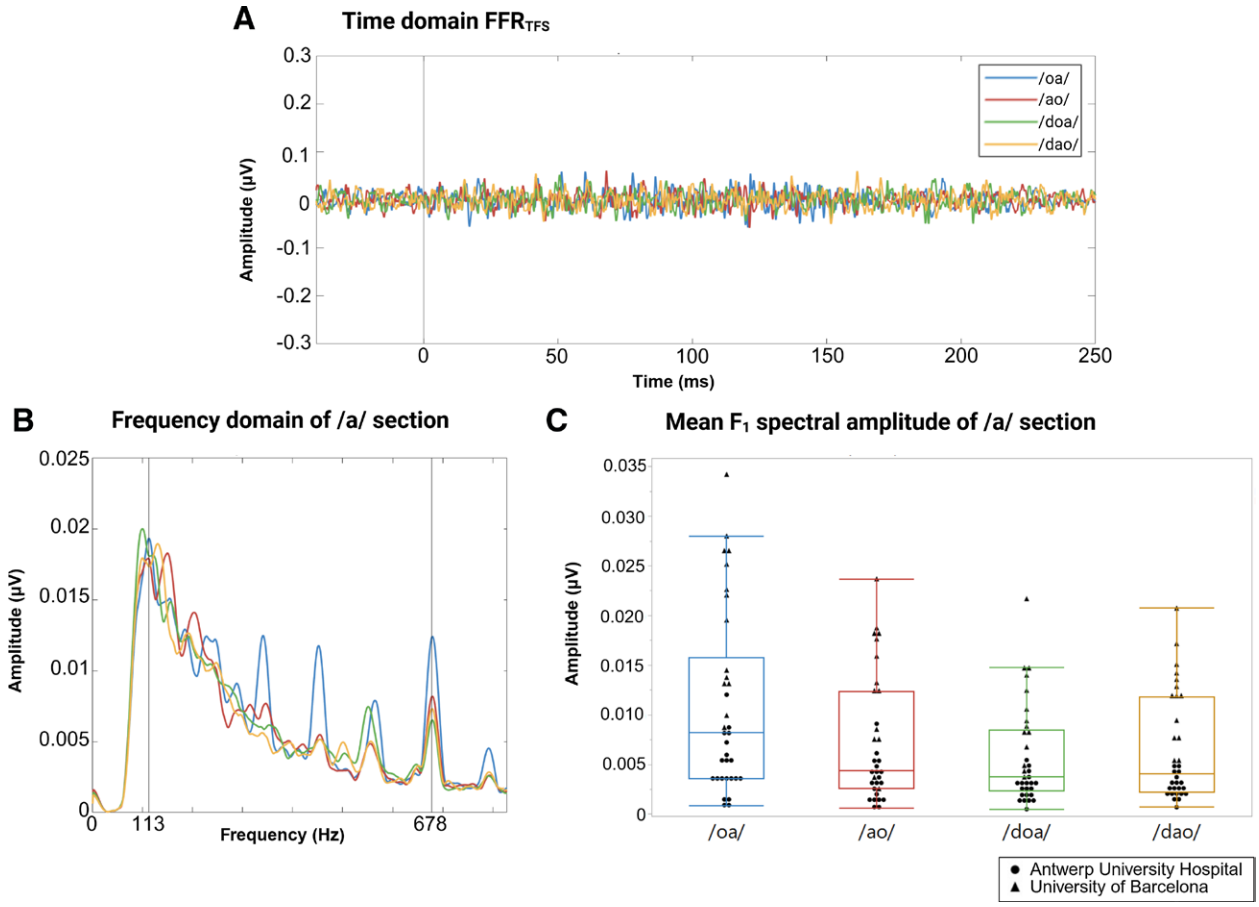

Fig. 4. Results for the FFR_{TFS} for /a/ section. A, Grand-averaged time-domain waveform of the FFR_{TFS} elicited by the four different stimuli. B, FFR_{TFS} frequency spectra extracted from the equivalent /a/ analyzed section of the neural response. C, Boxplots of the mean F_1 spectral amplitude from the equivalent /a/ analyzed section of the neural response for each stimulus.

In Antwerp, the effect of stimulus was weaker but still significant ($F_{3,96.01} = 3.13$, $p = 0.030$). The /oa/ elicited significantly higher amplitudes than /doa/ ($p = 0.034$) and /dao/ ($p = 0.044$).

These results confirm that stimulus-driven differences in F_1 response amplitude were present in both centers but were more pronounced in Barcelona. A full summary of estimated marginal means and pairwise comparisons is provided in Section E in Supplemental Digital Content, https://links.lww.com/EANDH/B792.

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

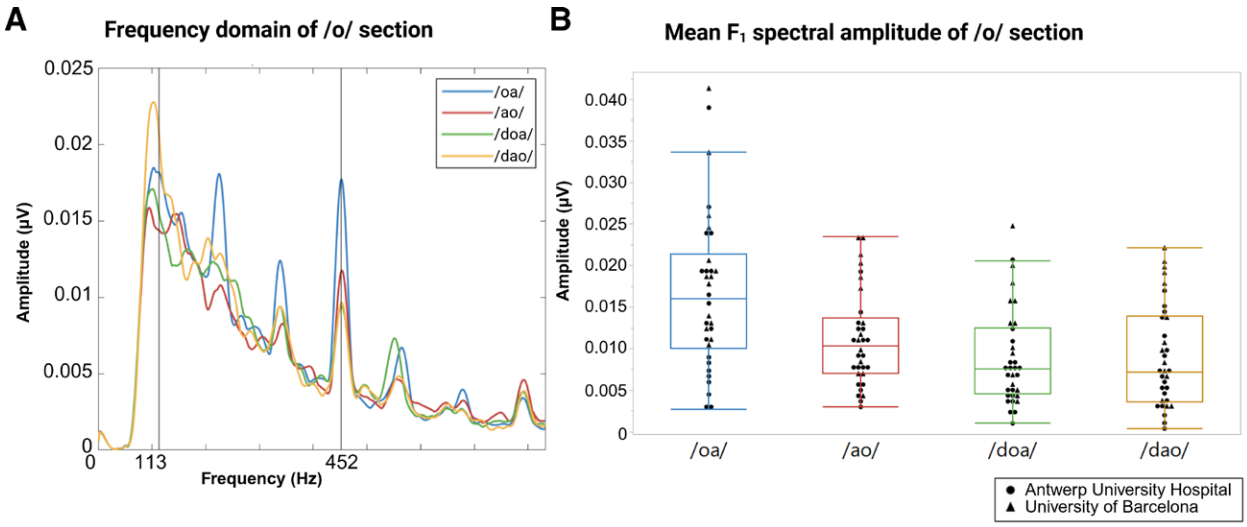

Fig. 5. Results for the FFR_{TFS} for /o/ section. A, FFR_{TFS} frequency spectra extracted from the equivalent /o/ analyzed section of the neural response. B, Boxplots of the mean F_1 spectral amplitude from the equivalent /o/ analyzed section of the neural response for each stimulus.

The SNR at F_1 for the /a/ section showed a significant stimulus effect ($F_{3,96} = 9.90$, $p < 0.001$). Post hoc comparisons indicated that /oa/ had significantly higher SNR values than /doa/ ($p < 0.001$), /dao/ ($p = 0.001$), and /ao/ ($p < 0.001$). SNR values were also significantly higher in Barcelona compared with Antwerp ($F_{1,32} = 30.30$, $p < 0.001$). No significant interaction was found ($F_{3,96} = 0.97$, $p = 1.000$).

F_1 Response – /o/ Section

Figure 5 presents the spectra and boxplots of F_1 spectral amplitudes for the /o/ section.

A significant main effect of stimulus was found on the F_1 response amplitude ($F_{3,96} = 20.81$, $p < 0.001$), with /oa/ eliciting significantly larger amplitudes than /doa/, /dao/, and /ao/ (all $p < 0.001$). There were no significant effects of the research center or interaction.

Similarly, the SNR at F_1 also showed a significant stimulus effect ($F_{3,96} = 9.70$, $p < 0.001$). Post hoc tests revealed that /oa/ elicited significantly higher SNR values than /doa/ ($p < 0.001$), /dao/ ($p < 0.001$), and /ao/ ($p = 0.048$). No significant effects of research center or interaction were found.

To visually summarize the main and interaction effects across all parameters, Figure 6 presents interaction plots of estimated marginal means for spectral amplitude and SNR at F_0 and F_1, for both /a/ and /o/ sections.

DISCUSSION

In this multisite study, we investigated how vowel order and consonant precedence influence neural responses to stimulus vowels as measured with the FFR. Linear mixed-effects analysis revealed a consistent and reproducible pattern in F_0 response amplitudes, with the largest F_0 response amplitude for the /ao/ stimulus and the smallest for the /oa/ stimulus for both the corresponding /a/ and /o/ sections of the stimuli. Conversely, the F_1 response amplitude exhibited an opposite pattern: the /oa/ stimulus evoked significantly higher F_1 amplitudes than all other stimuli in both vowel sections. We also observed a significantly longer neural lag for the /ao/ stimulus compared with the others. This may be due to its smoother acoustic onset, whereas the sharp burst of a preceding stop consonant like /d/ likely facilitates faster neural synchronization. Pre-stimulus RMS amplitude did not significantly differ between stimuli or research centers, indicating consistent baseline neural activity across conditions.

To confirm the presence of frequency-specific responses at F_0, we conducted an *F* test. Significant responses were observed in nearly all conditions at both the group and individual levels, supporting the robustness of phase-locked neural activity across stimuli. The only exception was the /o/ segment of the /oa/ stimulus, which did not reach significance at the group level and showed a lower proportion of significant responses at the individual level. This suggests that this specific segment may elicit weaker or more variable neural encoding of F_0.

It is important to note that these general stimulus-driven patterns were stable across both research centers. Stimulus × Research center interactions emerged only in a small subset of parameters—specifically, F_0 and F_1 amplitudes of the /a/ section—suggesting some site-specific variation in effect magnitude. Follow-up analyses confirmed that the effect of stimulus on F_0 and F_1 amplitudes was significantly stronger in Barcelona, where clearer and more consistent amplitude differences between stimuli were observed. In contrast, these effects were weaker and often non-significant in Antwerp. However, for most measures, including neural lag and pre-stimulus RMS, no interaction effects were observed, supporting the overall reproducibility of findings across research centers.

Differences in F_0 Response Amplitudes Between Different Stimuli

One possible explanation for the observed differences in F_0 response amplitude between stimuli is that the acoustic characteristics of the stimulus, such as vowel identity, may influence neural encoding of F_0. For example, the vowel /a/ has an F_1 at 678 Hz, while /o/ has a lower F_1 at 452 Hz. Although the mechanisms are not fully understood, it is possible that the

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

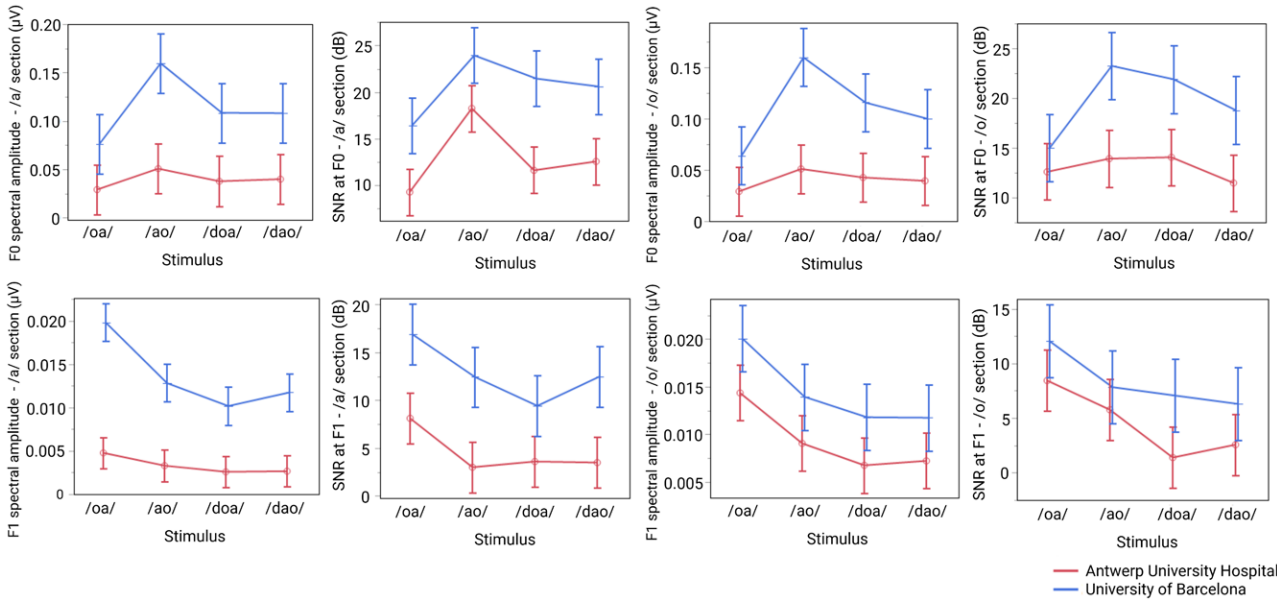

Fig. 6. Interaction plots of estimated marginal means for F_0 and F_1 spectral amplitudes and SNRs across stimuli and research centers. Each of the eight panels represents one FFR parameter: spectral amplitude and SNR at F_0 and F_1, separately for the /a/ and /o/ vowel segments. The x-axis indicates stimulus type (/doa/, /dao/, /oa/, /ao/), and the y-axis shows the adjusted mean. Each plotted point represents the estimated marginal mean derived from the statistical model. Vertical bars indicate the corresponding 95% confidence intervals. Values from Antwerp University Hospital are shown in red and values from the University of Barcelona in blue. Non-parallel lines indicate a significant interaction between stimulus and research center (e.g., for F_0 and F_1 amplitude of the /a/ section), while parallel lines indicate main effects without interaction.

vowel's spectral composition affects how strongly F_0 is encoded, potentially due to interactions in phase locking that begin at the level of the basilar membrane and continue along the auditory pathway. This could explain why F_0 amplitude in response to the /a/ section was consistently larger than that of the /o/ section across stimuli. The SNR results support this interpretation: stimuli with larger F_0 amplitudes, like /ao/, also showed higher SNRs, while stimuli with smaller amplitudes, like /oa/, had correspondingly lower SNRs. This pattern was observed consistently across both vowel segments.

In line with this, prior research has demonstrated that F_0 response amplitudes are affected by the frequency of the eliciting stimulus. Hoormann et al. (1992) recorded FFRs in response to 12 frequencies ranging from 128 to 832 Hz in 20 participants. Their data demonstrated that the grand-average FFR amplitude increased with frequency up to 320 to 384 Hz, after which it declined. This resembles the classical modulation transfer function observed in EFRs (Purcell et al. 2004).

This does not yet account for the discrepancies in F_0 response amplitude across the various stimuli, such as why the F_0 response amplitude to the /a/ section of the /ao/ stimulus is significantly larger than that of the /oa/ stimulus. One likely explanation is the influence of the preceding stimulus element, potentially through forward masking, a phenomenon in which a preceding sound suppresses the neural response to a closely following sound (Harris & Dallos, 1979). To illustrate, in the case of /oa/, the reduced excitation resulting from the /o/ section of the stimulus seems to persist in the /a/ section of the stimulus (Table 1). This would explain why the F_0 response amplitude to the /a/ section in /oa/ is significantly lower than that in /ao/.

When the consonant /d/ is added at the beginning of the stimulus, its F_1 frequency falls between those of /o/ and /a/. In the /dao/ stimulus, for instance, the F_1 starts at 553 to 565 Hz in the /d/ section, then rises to 678 Hz in the /a/ section, and then lowers again to 452 Hz in the /o/ section of the stimulus. When a stimulus begins with /d/, the F_0 response amplitude is therefore intermediate between that of /o/ and /a/. This explains why the F_0 response amplitudes in /doa/ and /dao/ are between those of /oa/ and /ao/ for both the corresponding /o/ and /a/ sections of the stimulus.

There are only a few studies investigating the influence of surrounding consonants and vowels on the FFR, and their findings have been inconsistent (Choi et al. 2013; Hodge et al. 2018; Easwar et al. 2021, 2022; Easwar & Chung 2022). Our results partly align with Easwar and colleagues' findings on how phonetic context influences the EFR. They observed that vowel identity significantly affects EFR amplitude, whereas consonant context does not (Easwar et al. 2021). Further experiments revealed that preceding phonemic context had no effect on EFRs elicited by the first formant (F_1), but certain consonants like /ʃ/ and /m/ attenuated EFRs for the second and higher formants (F_{2+}) under specific conditions (Easwar et al. 2022). In addition, adaptation effects showed that preceding phoneme sequences reduced EFR amplitude for F_{2+} only during the initial stimulus presentation, with no impact on F_1 (Easwar & Chung 2022). Differences across studies were linked to variations in context and stimulus characteristics (Easwar et al. 2022). Research on the effects of preceding stimuli on cortical auditory evoked potentials has shown that a preceding stimulus can reduce the amplitude of the following stimulus, particularly when the interstimulus interval is short (Wang et al. 2004). This reduction is most pronounced for the N1 response, which is a negative cortical AEP occurring around 100 msec after stimulus onset, when the preceding and following sound stimuli had the same

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

frequency (Butler 1968; Näätänen et al. 1988). This refractory effect can also be seen when the first stimulus has a different frequency from the second tone (Butler 1968; Näätänen et al. 1988).

It remains unclear whether these context effects originate from peripheral, subcortical, or cortical regions. Isolating the neural generators of the FFR was not the goal of our study, but it is an important direction for future research. Compelling magnetoencephalographic evidence indicates that FFRs elicited to F_0 in the range of 100 Hz, as in this study, have cortical sources (Coffey et al. 2016), whereas increasing F_0 to frequency ranges close to 300 Hz highlights subcortical contributions (midbrain and thalamic), making cortical sources silent (Gorina-Careta et al. 2021). However, to our knowledge, no studies so far have addressed the FFR sources elicited to higher frequencies, such as those of F_1 in the present stimuli. Recent work by Temboury-Gutierrez et al. (2024) showed that FFRs recorded at the scalp reflect peripheral auditory nerve activity, as evidenced by concurrent reductions in electrocochleographic and EEG responses in older adults. This supports the idea that peripheral neural degeneration can impact FFRs and highlights the potential of combining multi-channel EEG or ECochG with source analysis to further clarify the origins of context effects. In addition, future studies may benefit from the use of computational auditory models, such as the phenomenological auditory nerve model by Zilany et al. (2014), to evaluate whether the observed FFR effects can be predicted by peripheral auditory mechanisms alone. Thus, the approaches discussed earlier could offer further insight into the extent to which the effects reported here are shaped by early-stage auditory processing.

Differences in F_1 Response Amplitudes Between Different Stimuli

The differences observed in F_1 response amplitudes, and the discrepancy between F_0 and F_1 response amplitudes between the different eliciting stimuli, are difficult to explain and have not been addressed in existing literature. A possible hypothesis is that the F_0 amplitude may partially mask or obscure TFS amplitudes. This could account for why the /oa/ stimulus, which produces the smallest F_0 response amplitude, also generates the largest F_1 amplitude, as its F_0 response will mask the TFS less substantially. The presence of a preceding /d/, as is the case in /doa/ and /dao/, may mask the TFS more substantially, leading to lower F_1 response amplitudes. However, this hypothesis is not supported by existing literature, as no relevant studies were found. In addition, Post hoc analysis revealed that stimulus effects on F_1 amplitude were considerably more pronounced at the University of Barcelona than in Antwerp. This suggests that recording stimulus conditions can influence the detectability of stimulus-related effects in FFR_{TFS} responses, even when the overall trend remains consistent across sites.

Influence of Recording Conditions on F_0 and F_1 Response Amplitudes

The acquisition parameters differed between the two research centers. At Antwerp University Hospital, monaural stimulation at 70 dB SPL was used with IHS equipment, while at the University of Barcelona, stimuli were presented binaurally (diotically) at 75 dB SPL using Neuroscan. F_0 response amplitudes in the /a/ section were significantly smaller in Antwerp compared with Barcelona (all $p < 0.001$). For the /o/ section, significant differences were observed in three out of four stimuli, again with lower amplitudes in Antwerp. F_1 response amplitudes for /a/ were also lower in Antwerp, though no significant differences were observed for /o/.

These amplitude differences stem from both intensity and presentation mode. Binaural stimulation is known to elicit larger AEPs than monaural stimulation due to binaural summation (Butler et al. 1969; Krishnan & McDaniel 1998; Ballachanda & Moushegian 2000; Edmonds & Culling 2009; Lee et al. 2020). Moreover, AEPs show robust increases in amplitude with stimulus intensity (Camposano & Lolas 1992; Billings et al. 2007; Skoe & Kraus 2010).

Despite differences in stimulus presentation and acquisition setups between sites, the consistency in the direction of effects across stimuli and research centers underscores the robustness of our results. It is important to note that our linear mixed-effects models included research center as a factor and tested for Stimulus × Research center interactions, which were non-significant for most parameters. This confirms that although the acquisition setup influenced overall response amplitudes, it also modulated the strength of stimulus-related effects in specific cases, particularly for F_0 and F_1 amplitudes in the /a/ section. Stimulus-related amplitude differences were more reliably observed in Barcelona, likely due to the higher presentation level and binaural stimulation used at that site. However, non-linear effects in binaural summation and flooring effects in Antwerp recordings may also have contributed to these differences.

Implications

On the basis of our results, the preferred stimulus depends on the goal of the study. More specifically, the /ao/ stimulus might seem preferable for future studies focusing on the FFR_{ENV}, as it produces the largest F_0 response amplitude for both the /a/ and /o/ sections. If the focus of the study is to further investigate the FFR_{TFS}, which could be the case in developmental studies, for example, the /oa/ stimulus might be preferred because of the large F_1 response amplitudes it generates. If the goal is to investigate both the FFR_{ENV} and FFR_{TFS}, the choice of the stimulus is slightly more complex. The /dao/ stimulus offers a key advantage, as it incorporates the /da/ stimulus, allowing for separate analysis of the /da/ section. This feature enhances comparability with prior studies that have used the /da/ stimulus. Given this advantage, along with the fact that the /dao/ stimulus provided reliable responses for both the FFR_{ENV} and FFR_{TFS} in our study, we believe the /dao/ stimulus is an excellent choice for studies aiming to examine both the FFR_{ENV} and FFR_{TFS}.

While we are confident in our findings, we recognize several limitations of our study. One such limitation is the inclusion of participants with different language backgrounds, which may have influenced the results, as there is some evidence suggesting that language familiarity can impact the quality of the FFR signal during sound encoding (Skoe et al. 2022). However, in both centers, similar effects of stimulus variations on response amplitudes were seen. In addition, while all participants had clinically normal hearing and were within an adult age range, we cannot fully rule out subtle effects of subclinical age-related auditory processing decline or undetected variations in hearing sensitivity. Previous research has shown that both age and

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

hearing loss can affect FFRs, potentially influencing the neural representation of sound (Clinard et al. 2010; Parthasarathy et al. 2014; Jacxsens et al. 2024). Furthermore, although the /oa/ stimulus was originally designed for pediatric populations, the present study was conducted in adults. Future research will be necessary to confirm whether the patterns observed here also generalize to infants and children.

CONCLUSION

Our findings imply that the differences in F_0 response amplitudes may be influenced by the frequency of the eliciting vowel and the effects of preceding vowels and consonants in the stimulus. This underscores the impact of both vowel order and coarticulation on speech encoding strength. The choice of stimulus should be guided by the specific goals of the study. The /ao/ stimulus is ideal for studies focusing on the FFR_{ENV}. For studies targeting the FFR_{TFS}, the /oa/ stimulus may be a suitable option. For investigations examining both the FFR_{ENV} and FFR_{TFS}, we recommend the /dao/ stimulus, as it consistently generates strong responses and includes the widely used /da/ stimulus, enhancing comparability with prior research. While the overall pattern of stimulus effects was consistent across sites, these effects were more robust in Barcelona, emphasizing the potential influence of recording parameters on response magnitude.

ACKNOWLEDGMENTS

The authors acknowledge that this research was partially conducted and made possible through a research stay at the University of Barcelona, supported by a grant for a short research stay provided by the FWO (Fonds voor Wetenschappelijk Onderzoek—Research Foundation Flanders), grant number K208323N. The authors thank Vikram Norton for his help during the data collection phase.

C.E. was supported by the Spanish Ministry of Science and Innovation, the project PID2021-122255NB-100 supported by MCIN/AEI/10.13039/501100011033/FEDER, UE; the María de Maeztu Center of Excellence CEX2021-001159-M (supported by MCIN/AEI/10.13039/501100011033); the 2021SGR-00356 Consolidated Research Group of the Catalan Government, and the ICREA Acadèmia Distinguished Professorship award.

L.J. contributed to the concept, design, literature search, data acquisition, data analysis, and manuscript writing. N.G. contributed to the concept, design, data acquisition, data analysis, and manuscript editing and review. A.M. contributed to the concept, design, data acquisition, data analysis, and manuscript editing and review. S.A. contributed to the concept, design, and manuscript editing and review. L.B. contributed to data acquisition and manuscript editing and review. E.C. contributed to data analysis and manuscript editing and review. V.V.R. provided supervision and contributed to manuscript editing and review. W.D.H. contributed to the concept, design, supervision, and manuscript editing and review. M.L. contributed to the concept, design, supervision, and manuscript editing and review. C.E. contributed to the concept, design, supervision, and manuscript editing and review. All authors read and approved the final manuscript.

The datasets used and /or analyzed during the present study are available from the corresponding author on reasonable request.

The authors have no conflicts of interest to disclose.

Address for correspondence: Carles Escera, Brainlab—Cognitive Neuroscience Research Group, Department of Clinical Psychology and Psychobiology, University of Barcelona, Passeig Vall d'Hebron 171, Barcelona, 08035, Spain. E-mail: cescera@ub.edu; Laura Jacxsens, Department of Otorhinolaryngology, Head and Neck Surgery, Antwerp University Hospital, Drie Eikenstraat 655, Edegem, 2650, Belgium. E-mail: laura.jacxsens@uantwerpen.be

Received March 06, 2025; accepted September 28, 2025

REFERENCES

Aiken, S. J., & Picton, T. W. (2008). Envelope and spectral frequency-following responses to vowel sounds. *Hear Res*, *245*, 35–47.

Anderson, S., Parbery-Clark, A., White-Schwoch, T., Kraus, N. (2015). Development of subcortical speech representation in human infants. *J Acoust Soc Am*, *137*, 3346–3355.

Arenillas-Alcón, S., Costa-Faidella, J., Ribas-Prats, T., Gómez-Roig, M. D., Escera, C. (2021). Neural encoding of voice pitch and formant structure at birth as revealed by frequency-following responses. *Sci Rep*, *11*, 6660.

Arenillas-Alcón, S., Ribas-Prats, T., Puertollano, M., Mondéjar-Segovia, A., Gómez-Roig, M. D., Costa-Faidella, J., Escera, C. (2023). Prenatal daily musical exposure is associated with enhanced neural representation of speech fundamental frequency: Evidence from neonatal frequency-following responses. *Dev Sci*, *26*, e13362.

Ballachanda, B. B., & Moushegian, G. (2000). Frequency-following response: Effects of interaural time and intensity differences. *J Am Acad Audiol*, *11*, 1–11.

Bidelman, G. M., Gandour, J. T., Krishnan, A. (2011). Cross-domain effects of music and language experience on the representation of pitch in the human auditory brainstem. *J Cogn Neurosci*, *23*, 425–434.

Billings, C. J., Tremblay, K. L., Souza, P. E., Binns, M. A. (2007). Effects of hearing aid amplification and stimulus intensity on cortical auditory evoked potentials. *Audiol Neurootol*, *12*, 234–246.

Boersma, P., Weenink, D. (2013). *Praat: Doing Phonetics by Computer (Version 5.3.51*. https://praat.org/.

Bruckert, L., Thompson, P. A., Watkins, K. E., Bishop, D. V. M., Woodhead, Z. V. J. (2021). Investigating the effects of handedness on the consistency of lateralization for speech production and semantic processing tasks using functional transcranial Doppler sonography. *Laterality*, *26*, 680–705.

Butler, R. A. (1968). Effect of changes in stimulus frequency and intensity on habituation of the human vertex potential. *J Acoust Soc Am*, *44*, 945–950.

Butler, R. A., Keidel, W. D., Spreng, M. (1969). An investigation of the human cortical evoked potential under conditions of monaural and binaural stimulation. *Acta Otolaryngol*, *68*, 317–326.

Camposano, S., & Lolas, F. (1992). Effects of stimulation intensity, gender and handedness upon auditory evoked potentials. *Arq Neuropsiquiatr*, *50*, 43–49.

Choi, J. M., Purcell, D. W., Coyne, J. A., Aiken, S. J. (2013). Envelope following responses elicited by English sentences. *Ear Hear*, *34*, 637–650.

Clinard, C. G., Tremblay, K. L., Krishnan, A. R. (2010). Aging alters the perception and physiological representation of frequency: Evidence from human frequency-following response recordings. *Hear Res*, *264*, 48–55.

Coffey, E. B., Herholz, S. C., Chepesiuk, A. M., Baillet, S., Zatorre, R. J. (2016). Cortical contributions to the auditory frequency-following response revealed by MEG. *Nat Commun*, *7*, 11070.

Coffey, E. B. J., Nicol, T., White-Schwoch, T., Chandrasekaran, B., Krizman, J., Skoe, E., Zatorre, R. J., Kraus, N. (2019). Evolving perspectives on the sources of the frequency-following response. *Nat Commun*, *10*, 5036.

Delorme, A., & Makeig, S. (2004). EEGLAB: An open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *J Neurosci Methods*, *134*, 9–21.

Dobie, R. A., & Wilson, M. J. (1996). A comparison of t test, F test, and coherence methods of detecting steady-state auditory-evoked potentials, distortion-product otoacoustic emissions, or other sinusoids. *J Acoust Soc Am*, *100*, 2236–2246.

Easwar, V., & Chung, L. (2022). The influence of phoneme contexts on adaptation in vowel-evoked envelope following responses. *Eur J Neurosci*, *56*, 4572–4582.

Easwar, V., Bridgwater, E., Purcell, D. (2021). The influence of vowel identity, vowel production variability, and consonant environment on envelope following responses. *Ear Hear*, *42*, 662–672.

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.

Easwar, V., Boothalingam, S., Wilson, E. (2022). Sensitivity of vowel-evoked envelope following responses to spectra and level of preceding phoneme context. *Ear Hear*, *43*, 1327–1335.

Edmonds, B. A., & Culling, J. F. (2009). Interaural correlation and the binaural summation of loudness. *J Acoust Soc Am*, *125*, 3865–3870.

Encina-Llamas, G., Harte, J. M., Dau, T., Shinn-Cunningham, B., Epp, B. (2019). Investigating the effect of cochlear synaptopathy on envelope following responses using a model of the auditory nerve. *J Assoc Res Otolaryngol*, *20*, 363–382.

Fleming, D., Belleville, S., Peretz, I., West, G., Zendel, B. R. (2019). The effects of short-term musical training on the neural processing of speech-in-noise in older adults. *Brain Cogn*, *136*, 103592.

Font-Alaminos, M., Ribas-Prats, T., Gorina-Careta, N., Escera, C. (2021). Emergence of prediction error along the human auditory hierarchy. *Hear Res*, *399*, 107954.

Gorina-Careta, N., Arenillas-Alcón, S., Puertollano, M., Mondéjar-Segovia, A., Ijjou-Kadiri, S., Costa-Faidella, J., Gómez-Roig, M. D., Escera, C. (2024). Exposure to bilingual or monolingual maternal speech during pregnancy affects the neurophysiological encoding of speech sounds in neonates differently. *Front Hum Neurosci*, *18*, 1379660.

Gorina-Careta, N., Kurkela, J. L. O., Hämäläinen, J., Astikainen, P., Escera, C. (2021). Neural generators of the frequency-following response elicited to stimuli of low and high frequency: A magnetoencephalographic (MEG) study. *Neuroimage*, *231*, 117866.

Gorina-Careta, N., Ribas-Prats, T., Arenillas-Alcón, S., Puertollano, M., Gómez-Roig, M. D., Escera, C. (2022). Neonatal frequency-following responses: A methodological framework for clinical applications. *Semin Hear*, *43*, 162–176.

Harris, D. M., & Dallos, P. (1979). Forward masking of auditory nerve fiber responses. *J Neurophysiol*, *42*, 1083–1107.

Hodge, S. E., Menezes, D. C., Brown, K. D., Grose, J. H. (2018). Forward masking of the speech-evoked auditory brainstem response. *Otol Neurotol*, *39*, 150–157.

Hoormann, J., Falkenstein, M., Hohnsbein, J., Blanke, L. (1992). The human frequency-following response (FFR): Normal variability and relation to the click-evoked brainstem response. *Hear Res*, *59*, 179–188.

Jacxsens, L., Biot, L., Escera, C., Gilles, A., Cardon, E., Van Rompaey, V., De Hertogh, W., Lammers, M. J. W. (2024). Frequency-following responses in sensorineural hearing loss: A systematic review. *J Assoc Res Otolaryngol*, *25*, 131–147.

Kraus, N., & Chandrasekaran, B. (2010). Music training for the development of auditory skills. *Nat Rev Neurosci*, *11*, 599–605.

Kraus, N., & White-Schwoch, T. (2015). Unraveling the biology of auditory learning: A cognitive-sensorimotor-reward framework. *Trends Cogn Sci*, *19*, 642–654.

Kraus, N., McGee, T. J., Carrell, T. D., Zecker, S. G., Nicol, T. G., Koch, D. B. (1996). Auditory neurophysiologic responses and discrimination deficits in children with learning problems. *Science*, *273*, 971–973.

Krishnan, A., & McDaniel, S. S. (1998). Binaural interaction in the human frequency-following response: Effects of interaural intensity difference. *Audiol Neurootol*, *3*, 291–299.

Krishnan, A., Xu, Y., Gandour, J., Cariani, P. (2005). Encoding of pitch in the human brainstem is sensitive to language experience. *Brain Res Cogn Brain Res*, *25*, 161–168.

Krizman, J., & Kraus, N. (2019). Analyzing the FFR: A tutorial for decoding the richness of auditory function. *Hear Res*, *382*, 107779.

Lee, J. Y., Kang, B. C., Park, J. W., Park, H. J. (2020). Changes in cortical auditory evoked potentials by ipsilateral, contralateral and binaural speech stimulation in normal-hearing adults. *Clin Exp Otorhinolaryngol*, *13*, 133–140.

Lemos, F., Nunes, A., Evangelista, C., Escera, C., Taveira, K., Balen, S. (2021). Frequency-following response in newborns and infants: A systematic review of acquisition parameters. *J Speech Lang Hear Res*, *64*, 1–18.

Liu, F., Maggu, A. R., Lau, J. C. Y., Wong, P. C. M. (2015). Brainstem encoding of speech and musical stimuli in congenital amusia: Evidence from Cantonese speakers. *Front Hum Neurosci*, *8*, 1029.

Maddieson, I. (1984). *Patterns of sounds*. (Vol. *2*). Cambridge University Press.

Musacchia, G., Sams, M., Skoe, E., Kraus, N. (2007). Musicians have enhanced subcortical auditory and audiovisual processing of speech and music. *Proc Natl Acad Sci U S A*, *104*, 15894–15898.

Näätänen, R., Sams, M., Alho, K., Paavilainen, P., Reinikainen, K., Sokolov, E. N. (1988). Frequency and location specificity of the human vertex N1 wave. *Electroencephalogr Clin Neurophysiol*, *69*, 523–531.

Parthasarathy, A., Datta, J., Torres, J. A., Hopkins, C., Bartlett, E. L. (2014). Age-related changes in the relationship between auditory brainstem responses and envelope-following responses. *J Assoc Res Otolaryngol*, *15*, 649–661.

Picton, T., & Burkard, R. (2012). Human auditory evoked potentials. *J Acoust Soc Am*, *132*, 566.

Purcell, D. W., John, S. M., Schneider, B. A., Picton, T. W. (2004). Human temporal auditory acuity as assessed by envelope following responses. *J Acoust Soc Am*, *116*, 3581–3593.

Ribas-Prats, T., Almeida, L., Costa-Faidella, J., Plana, M., Corral, M. J., Gómez-Roig, M. D., Escera, C. (2019). The frequency-following response (FFR) to speech stimuli: A normative dataset in healthy newborns. *Hear Res*, *371*, 28–39.

Ribas-Prats, T., Arenillas-Alcón, S., Lip-Sosa, D. L., Costa-Faidella, J., Mazarico, E., Gómez-Roig, M. D., Escera, C. (2022). Deficient neural encoding of speech sounds in term neonates born after fetal growth restriction. *Dev Sci*, *25*, e13189.

Skoe, E., García-Sierra, A., Ramírez-Esparza, N., Jiang, S. (2022). Automatic sound encoding is sensitive to language familiarity: Evidence from English monolinguals and Spanish-English bilinguals. *Neurosci Lett*, *777*, 136582.

Skoe, E., & Kraus, N. (2010). Auditory brain stem response to complex sounds: A tutorial. *Ear Hear*, *31*, 302–324.

Song, J. H., Skoe, E., Wong, P. C., Kraus, N. (2008). Plasticity in the adult human auditory brainstem following short-term linguistic training. *J Cogn Neurosci*, *20*, 1892–1902.

Temboury-Gutierrez, M., Märcher-Rørsted, J., Bille, M., Yde, J., Encina-Llamas, G., Hjortkjær, J., Dau, T. (2024). Electrocochleographic frequency-following responses as a potential marker of age-related cochlear neural degeneration. *Hear Res*, *446*, 109005.

Turner, C. W., Fabry, D. A., Barrett, S., Horwitz, A. R. (1992). Detection and recognition of stop consonants by normal-hearing and hearing-impaired listeners. *J Speech Hear Res*, *35*, 942–949.

Copyright © 2025 The American Auditory Society. Unauthorized reproduction of this article is prohibited.